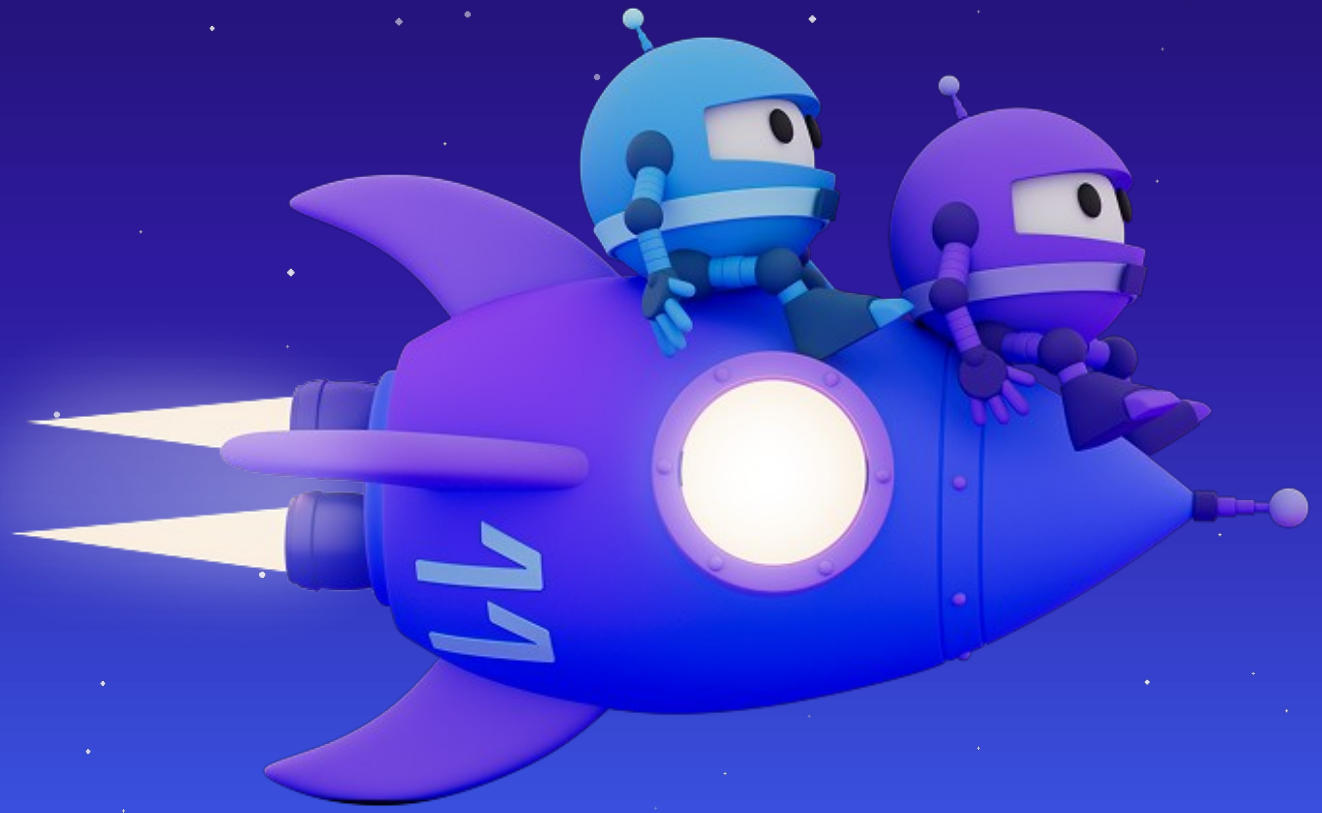
.NET + Foundry: Better Together



Bruno Capuano

Principal Cloud Advocate



.NET gives the application shape



Foundry brings the AI capabilities

MEAI and MAF serve different layers

MEAI for model access; MAF for agent behavior

MICROSOFT AGENT FRAMEWORK

AIAgent

Base class for agents with instructions, tools, sessions and workflows

Orchestration • tools • state



An agent can use IChatClient, but not every custom AIAgent must

MICROSOFT.EXTENSIONS.AI

IChatClient

Interface for one chat contract across compatible model providers

Messages • responses • middleware

Compatible model providers

Foundry supports three .NET paths



Model calls, app-owned agents and service-managed agents

DIRECT MODEL CALL

IChatClient	→	Foundry model deployment

APP-OWNED AGENT

AIAgent in your process	→	Foundry model deployment + tools

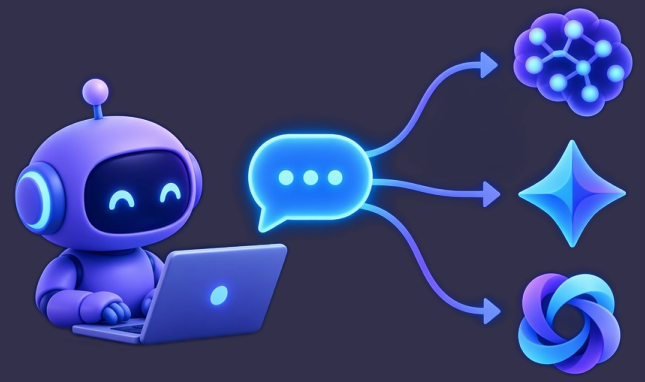
SERVICE-MANAGED AGENT

.NET client	→	Foundry prompt or hosted agent

Your code can call Foundry services without moving the agent runtime into Foundry

One call does not mean identical support

Keep the app contract stable, verify every deployment



Stable app contract

One prompt
One IChatClient call
Same application flow
Compare one question

Deployment reality

Model availability
Responses and tool support
Output behavior
Verify before the demo

Consuming is not deploying

Two agent paths

Prompt agent

Model + instructions + tools

Consume an existing endpoint or definition

Hosted code

Custom runtime + protocol

Package and deploy custom agent code



A hosted agent can also be consumed, and prompt definitions also have creation and versioning lifecycles

Where should the agent code run?

Choose the operating model first

MANAGED AGENT RUNTIME

Foundry hosted agents

- Platform runs the agent container
- Built-in endpoint and agent identity
- Managed scaling and lifecycle

Best when: agent-specific managed hosting

MANAGED CONTAINER PLATFORM

Azure Container Apps

- APIs, workers and microservices
- Ingress, revisions and event-driven scaling
- No direct Kubernetes API administration

Best when: container flexibility with less platform work

KUBERNETES PLATFORM

Azure Kubernetes Service

- Direct Kubernetes API
- Cluster-level workload control
- AKS Automatic can reduce operations

Best when: platform teams need Kubernetes control

Where the code runs is separate from which Foundry models and tools it calls

Sources: Microsoft Learn, Hosted agents in Foundry Agent Service; Compare Azure container options with AKS

Reliable agents need boundaries

CONTROL

Toolboxes

Give tools narrow purpose and permissions

Guardrails

Validate inputs, outputs and actions

CONTEXT

Memory

Keep only the context the experience needs

Identity

AzureCliCredential locally
DefaultAzureCredential when hosted

OBSERVE AND VERIFY

Tracing

Inspect model calls, tool calls and failures

Provider fit

Verify model, Responses and tool support per deployment

Keep capabilities narrow and dependencies observable

Thank you

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